

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

I **T.Ganesh(192425246)** (Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity

1.WLAN + WWAN Performance & Security Issues (Smart Office Building)

Problem Understanding More than 1,200 employees rely on Wi-Fi (WLAN) inside a 6-storey building and field staff use 4G/5G (WWAN). Users face:

- Slow speed during peak hours
- Frequent disconnections while moving between floors
- Weak signal in conference rooms
- Video calls freezing
- Uneven load on access points (some overloaded, some idle)
- Field engineers losing connectivity in rural/highway areas
- Some devices still using outdated security protocols

Problem Reported

(Slow speed, disconnections, weak signal, security issues)



Collect Data

(AP load, signal strength, roaming logs, security protocols)



Identify Root Causes

(Poor AP placement, no roaming, interference, old security)



Debug the Network

(Site survey, spectrum analysis, security audit)



Apply Optimisation

(Add APs, enable fast roaming, WPA3, load balancing, dual-SIM)



Final Result

(Stable, secure and even wireless connectivity)

Root Causes

- Poor wireless site survey → insufficient AP density and wrong channel planning
- No seamless roaming (missing 802.11r/k/v)
- Channel interference and co-channel congestion
- No load balancing / band steering
- Overloaded 2.4 GHz band
- Weak coverage in conference rooms
- WWAN: no dual-SIM/failover or signal boosters
- Legacy security (WPA/WPA2-TKIP or open SSIDs still active)

Debugging Steps

1. Perform a professional wireless site survey (Ekahau / AirMagnet) – measure RSSI, SNR, channel utilisation.
2. Check AP client count and CPU utilisation.
3. Analyse spectrum for interference (microwave, Bluetooth, neighbouring Wi-Fi).
4. Verify roaming logs and 802.11r/k/v status.
5. Audit security: list all SSIDs and encryption methods.
6. For WWAN: check signal strength (RSRP/RSRQ) and carrier coverage maps.

Optimisation Recommendations

- Redesign WLAN with proper AP density (one AP per 15–25 users) and dual-band (prefer 5 GHz / 6 GHz).
- Enable 802.11r (Fast Transition), 802.11k and 802.11v for seamless roaming.
- Implement band steering and client load balancing.
- Add directional APs or extra APs in conference rooms.
- Upgrade all SSIDs to WPA3-Enterprise (or at least WPA2-AES).
- For WWAN: deploy dual-SIM routers with automatic failover + external antennas/signal boosters for field engineers.
- Enable QoS (WMM + DSCP marking) for voice/video traffic.

- Schedule automatic RF optimisation (Cisco RRM / Aruba ARM).

Expected Result Stable roaming, even load distribution, strong signal everywhere, secure connections, and reliable remote access for field staff.

2. ATM Network Performance Issues (Financial Organisation)

Problem Understanding ATM network carrying online banking, VoIP, video and real-time transactions is suffering from:

- Transaction delays
- Voice jitter and packet loss
- Poor video quality
- High utilisation on some ATM switches
- Congested Virtual Circuits (VCs)
- Wrong ATM service categories being used

Root Causes

- Incorrect ATM service category assignment (e.g., using UBR for voice instead of CBR/rt-VBR)
- No proper traffic shaping / policing
- Overloaded VCs and lack of bandwidth reservation
- Missing or poorly configured QoS parameters (PCR, SCR, MBS)

- Congestion at ATM switches without adequate buffering or ABR mechanisms

Debugging Steps

1. Map every application to its VC and service category.
2. Check cell loss ratio, cell delay variation (CDV) and cell transfer delay on each VC.
3. Monitor switch buffer utilisation and cell discard counters.
4. Verify Peak Cell Rate (PCR), Sustainable Cell Rate (SCR) and Maximum Burst Size (MBS) values.
5. Identify which VCs are oversubscribed.

Optimisation Recommendations

- Reclassify traffic correctly:
 - Voice → CBR (Constant Bit Rate)
 - Video → rt-VBR (real-time Variable Bit Rate)
 - Transactions → nrt-VBR or ABR
 - Bulk data → UBR or GFR
- Apply proper traffic policing and shaping at the edge.
- Increase bandwidth on congested VCs or create additional parallel VCs.
- Enable Early Packet Discard (EPD) / Partial Packet Discard (PPD) to reduce cell loss impact.
- Implement Connection Admission Control (CAC) so new VCs are only accepted if resources exist.

- Consider gradual migration to MPLS or IP/MPLS if ATM is becoming legacy.

Expected Result Predictable low-latency for banking transactions, clear voice, stable video, and efficient bandwidth use.

3. MPLS Network Congestion & Incorrect Label Mapping

Problem Understanding MPLS backbone connecting HQ, regional offices, warehouses and cloud data centres is experiencing:

- Slow cloud access and delayed order processing
- Poor VoIP and video quality
- Some LSPs heavily congested, others under-utilised
- Incorrect label mappings on routers
- No proper traffic engineering

Root Causes

- Static or poorly designed Label Switched Paths (LSPs)
- Missing or incorrect LDP / RSVP-TE label bindings
- No Traffic Engineering (TE) tunnels or constraint-based routing
- Uneven IGP metrics causing suboptimal path selection
- Lack of Fast Reroute (FRR) or backup LSPs

Debugging Steps

1. Check label bindings (show mpls ldp bindings / show mpls forwarding-table).

2. Trace LSPs end-to-end and measure latency/jitter/packet loss.
3. Identify congested links using interface utilisation and MPLS TE tunnel statistics.
4. Verify IGP (OSPF/IS-IS) metrics and TE database.
5. Look for label mismatches or blackholing.

Optimisation Recommendations

- Deploy MPLS Traffic Engineering (RSVP-TE) with explicit path tunnels for critical traffic.
- Create separate TE tunnels for Voice, Video, ERP and best-effort traffic.
- Enable Auto-Bandwidth and Bandwidth Protection.
- Correct all label mappings and enable LDP session protection.
- Implement Fast Reroute (Link and Node protection) for sub-50 ms recovery.
- Use Differentiated Services (DiffServ-aware TE) so high-priority traffic gets preferential treatment.
- Monitor with MPLS OAM (LSP ping/traceroute) and NetFlow/IPFIX.

Expected Result Evenly utilised backbone, low latency for critical applications, fast recovery from failures, and improved customer experience.

4. Transmission Errors in Automated Warehouse Network

Problem Understanding Barcode scanners, RFID readers, robots and inventory systems exchange data continuously.

Problems observed:

- Incorrect shipment records
- Failed barcode scans
- Inventory mismatch
- Increased transmission errors due to electrical interference, attenuation and congestion
- Some corrupted frames go undetected

Root Causes

- Weak or missing error detection (CRC not properly verified or only parity used)
- No or inadequate Forward Error Correction (FEC)
- Electrical noise from motors/robots affecting copper cabling
- Signal attenuation on long runs
- Network congestion causing collisions or buffer overflows
- Lack of end-to-end integrity checks at application layer

Debugging Steps

1. Capture frames and check FCS/CRC error counters on switches and NICs.
2. Measure cable quality (TDR test) and check for EMI sources.

3. Monitor collision and late-collision statistics (if half-duplex still exists).
4. Verify whether higher-layer protocols (TCP) are detecting the remaining errors.
5. Check RFID/barcode device firmware and signal strength.

Optimisation Recommendations

- Move to full-duplex switched Ethernet everywhere (eliminate collisions).
- Use shielded cabling (S/FTP) or fibre where EMI is high.
- Ensure strong CRC-32 is enabled and monitored.
- Add application-layer checksums or digital signatures for critical inventory records.
- Implement Forward Error Correction (FEC) on links that still suffer residual errors.
- Use QoS to prioritise scanner/RFID traffic and prevent congestion drops.
- Deploy industrial-grade switches with better noise immunity.
- Schedule regular cable and connector inspection.

Expected Result Near-zero undetected errors, accurate inventory data, reliable barcode/RFID operation even during peak hours.

5. Flow Control Problems in Online Education Platform

Problem Understanding Central server (10 Gbps) sends exam papers, images, videos and acknowledgements to 15,000+ students on slower broadband/mobile links. Issues:

- Incomplete downloads
- Delayed page loading
- Duplicate packets and excessive retransmissions
- Receiver buffer overflow
- High latency and poor bandwidth utilisation

Root Causes

- Classic mismatch between fast sender and slow receivers (no effective flow control)
- TCP window too large or not adapting properly
- Lack of congestion control awareness for heterogeneous receivers
- No traffic shaping or rate limiting at the server
- Receiver buffers too small for the offered load
- Possible use of UDP without application-level flow control for some media

Debugging Steps

1. Analyse TCP window size, retransmission rate and zero-window advertisements.
2. Measure round-trip time (RTT) and bandwidth-delay product for typical student links.

3. Check server-side send buffer and student-side receive buffer sizes.
4. Look for packet loss vs. buffer overflow symptoms.
5. Capture traffic during peak exam time to see burst behaviour.

Optimisation Recommendations

- Implement proper TCP flow control and modern congestion control (BBR or CUBIC).
- Use Adaptive Bitrate (ABR) streaming for videos and progressive download for large files.
- Deploy a Content Delivery Network (CDN) or edge caching so students fetch from nearby servers.
- Apply server-side rate limiting / traffic shaping per student or per class.
- Increase receive buffers on student devices where possible and use TCP window scaling.
- For real-time parts, use WebRTC or QUIC which handle flow control better than classic TCP.
- Implement application-level acknowledgements and pacing so the server does not overwhelm slow clients.
- Monitor with tools that show per-flow throughput and retransmission ratio.

Expected Result Smooth exam delivery, minimal retransmissions, no buffer overflows, fair bandwidth sharing, and stable experience even for students on slow mobile links.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation